Avik Kumar

Atlanta, GA | akumar3076@gatech.edu | LinkedIn | GitHub

Education

Georgia Institute of Technology Atlanta, GA

Bachelor of Science in Computer Science — Intelligence, Modeling & Simulation *Expected May 2028*

GPA: 4.00

Relevant Coursework: Intro to AI, Algorithm Analysis & Design, Data Structures, Discrete Math, Computer Organization, Systems & Networks

Experience

Georgia Tech Solar Racing Atlanta, GA

Data Subteam Lead *Aug 2025 – Present*

- Promoted to lead in August 2026; lead a **10+ member** software team across simulation, strategy, and real-time visualization for the team's solar car.
- Built and deployed the team's real-time telemetry visualizer with **AWS EC2, Grafana, and InfluxDB** for live vehicle tracking, lap recording, and telemetry analysis during FSGP and ASC (**1,500+ miles, 80+ hours**).
- Develop a closed-loop **MATLAB/Simulink** vehicle simulation coupling solar generation, battery discharge/voltage, and subsystem efficiencies to characterize losses and predict realistic vehicle performance.

Ha Lab, Georgia Tech Atlanta, GA

Robotics & Reinforcement Learning Research Assistant *Jan 2026 – Present*

- Develop teleoperation and data-collection infrastructure for training **flow-matching policies** for whole-body navigation and manipulation on the **Unitree G1** humanoid.
- Extend the **GEAR-SONIC** teleoperation stack across hardware and simulation, enabling synchronized whole-body and per-finger demonstrations for policy training.
- Develop **grasp and base-placement guidance priors** that condition policy actions toward feasible manipulation and navigation behaviors during autonomous execution.

ML for Logic @ Georgia Tech Atlanta, GA

SMT Solver Research Assistant *Nov 2025 – Present*

- Improve Monte Carlo Tree Search strategy synthesis for the **Z3 SMT solver** by replacing **UCT** action selection with **PUCT** using LLM-generated tactic priors.
- Develop **evolutionary feedback loops** that use solver performance to iteratively refine LLM-guided tactic sequences and synthesize targeted, high-performing solving strategies.
- Contributed LLM-guided tactic selection to a solver entered in **SMT-COMP 2026**, earning **first-place awards** across multiple participating tracks and logic divisions.

Projects

ArxivReads | *Python, Django, OpenAlex API* Feb 2026—Present

- Developing a Goodreads-style platform for discovering, organizing, and tracking academic papers, with **semantic search** for finding relevant research beyond keyword matching.
- Built personalized reading workflows for saving papers to lists and tracking 'To Read," 'Currently Reading," and "Read" statuses.

Second Brain | *Python, Discord API, Docker, Markdown* May 2026

- Built a **Discord-based** personal knowledge system for natural-language querying, task management, and persistent organization of projects and documentation.
- Designed a **specialized-agent architecture** with **PDF and text ingestion** to automatically turn Discord interactions and reference material into structured Markdown knowledge.

Technical Skills

Languages: Python, Java, HTML/CSS

ML & Robotics: NumPy, pandas, scikit-learn, MuJoCo, Gymnasium, Stable-Baselines3, Simulink

Systems & Infrastructure: Git, Docker, PostgreSQL, SQLite, AWS EC2, Grafana, InfluxDB

Frameworks & Tools: Django, Flask, React, Streamlit, Jupyter, Z3, MATLAB

Human Languages: English (native), Hindi (native), French (basic)